

Reflection of Light - HKDSE Physics Summary

1. Law of Reflection

1.1 The Law

When light reflects off a surface:

Law of Reflection:

$$i = r$$

Where:

- i
= angle of incidence (入射角)
- r
= angle of reflection (反射角)

Important Notes:

- Angles are measured from the **normal** (perpendicular to surface)
- NOT measured from the surface itself
- The law applies to **all types of reflection** (regular and diffuse)

1.2 Key Terms

- **Incident ray** (入射光線) : Light ray approaching the surface
 - **Reflected ray** (反射光線) : Light ray leaving the surface
 - **Normal** (法線) : Perpendicular line to the surface at point of incidence
 - **Angle of incidence** (入射角) : Angle between incident ray and normal
 - **Angle of reflection** (反射角) : Angle between reflected ray and normal
 - **Point of incidence** (入射點) : Point where light hits the surface
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2. Types of Reflection

2.1 Regular Reflection (Specular Reflection)

- Occurs on **smooth, polished surfaces** (mirrors, calm water, polished metal)
- Parallel incident rays produce **parallel reflected rays**

- Forms **clear images**
- All normals at different points are parallel

Examples:

- Plane mirror
- Polished metal surface
- Calm water surface

2.2 Diffuse Reflection (Irregular Reflection)

- Occurs on **rough, uneven surfaces** (paper, wall, clothing)
- Parallel incident rays produce **scattered reflected rays** in different directions
- Does **not** form clear images
- Normals at different points point in different directions

Examples:

- Paper
- Rough wall
- Clothing
- Unpolished wood

Important Note:

- The law of reflection ($i = r$) **still applies** at each point
 - The difference is due to the **surface roughness**, not the law itself
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3. Plane Mirror Image Formation

3.1 Image Characteristics

An image formed by a plane mirror has the following characteristics:

1.  **Virtual** - Cannot be projected on screen (formed by extensions of reflected rays)
2.  **Erect** - Same orientation as object (upright)
3.  **Same size** - Image height = Object height
4.  **Laterally inverted** - Left and right are reversed

5. ☒ **Image distance = Object distance** - Image is as far behind mirror as object is in front

Mathematical Expression:

$$v = u$$

Where:

- v
= image distance from mirror
- u
= object distance from mirror

3.2 Ray Diagram for Plane Mirror

To locate image position:

1. Draw at least **two rays** from object point
 2. Rays reflect following
 $i = r$
 3. Extend reflected rays **behind mirror** (dotted lines)
 4. Image is where extensions intersect
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4. Types of Light Beams

4.1 Parallel Beam

- All light rays are **parallel** to each other
- Travel in the **same direction**
- Do not converge or diverge

Sources:

- Distant light source (Sun, distant star)
- Searchlight
- Laser

4.2 Converging Beam

- Light rays **converge** (come together) at a point

- Rays travel towards a common point

Formation:

- After passing through convex lens
- Reflected from concave mirror

4.3 Diverging Beam

- Light rays **diverge** (spread out) from a point
- Rays appear to come from a common point

Formation:

- From a point source (light bulb)
 - After passing through concave lens
 - Reflected from convex mirror
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5. Applications of Reflection

5.1 Plane Mirror

- **Bathroom mirror:** Personal grooming
- **Rear-view mirror:** See behind vehicle
- **Periscope:** See over obstacles (uses two plane mirrors)
- **Kaleidoscope:** Create symmetrical patterns

5.2 Curved Mirrors

Concave Mirror (Converging):

- Shaving mirror (magnified image)
- Dentist's mirror
- Telescope mirror
- Solar concentrator

Convex Mirror (Diverging):

- Car side mirror (wider field of view)
- Security mirror in shops
- Road safety mirror at corners

6. Minimum Mirror Length

6.1 To See Full Body Image

For a person to see their full body in a plane mirror:

Minimum mirror length =

$$\frac{1}{2}$$

× Person's height

Reasoning:

- By geometry and law of reflection
- Mirror must extend from eye level to halfway between eyes and feet
- Mirror position matters: Top of mirror at midpoint between eyes and top of head

6.2 Mirror Position

- **Top of mirror:** Halfway between eyes and top of head
 - **Bottom of mirror:** Halfway between eyes and feet
 - Distance from mirror does **not** affect minimum length
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7. Multiple Reflections

7.1 Two Parallel Mirrors

- Produce **infinite images**
- Images alternate between left and right sides
- Each subsequent image is further away

7.2 Two Mirrors at Angle

Number of images formed:

$$N = \frac{360^\circ}{\theta} - 1$$

Where:

- N
= number of images

- θ
= angle between mirrors

Examples:

- $\theta = 90^\circ$
:
 $N = 3$
images
 - $\theta = 60^\circ$
:
 $N = 5$
images
 - $\theta = 45^\circ$
:
 $N = 7$
images
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8. Important Concepts

8.1 Virtual vs Real Image

Virtual Image (Plane Mirror):

- Formed by **extensions** of reflected rays
- Cannot be projected on screen
- Appears behind mirror
- Always erect

Real Image (Curved Mirrors):

- Formed by **actual convergence** of reflected rays
- Can be projected on screen
- Appears in front of mirror
- Usually inverted

8.2 Lateral Inversion

- Left and right are **reversed** in plane mirror
- Top and bottom remain the same

- Text appears reversed (mirror writing)
- Clock appears to run backwards

Example:

- Word "AMBULANCE" written reversed on front of ambulance
 - Can be read correctly in rear-view mirror
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9. Problem-Solving Strategies

9.1 Ray Diagram Method

1. Draw normal at point of incidence
2. Measure angle of incidence from normal
3. Apply
$$i = r$$
to find angle of reflection
4. Draw reflected ray at angle
$$r$$
from normal

9.2 Image Location Method (Plane Mirror)

1. Image distance = Object distance
2. Image is behind mirror
3. Draw at least two rays and extend behind mirror
4. Image is where extensions intersect

9.3 Calculation Tips

- Always measure angles from **normal**, not surface
 - Use geometry: Angles in triangle sum to 180°
 - Use symmetry: Image and object are symmetrical about mirror
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10. Common Misconceptions

10.1 Misconception 1

✗ **Wrong:** Angles measured from surface

✓ **Correct:** Angles measured from normal (perpendicular to surface)

10.2 Misconception 2

✗ **Wrong:** Diffuse reflection doesn't follow

$$i = r$$

✓ **Correct:** Law of reflection applies to all reflection; diffuse reflection is due to rough surface

10.3 Misconception 3

✗ **Wrong:** Image in plane mirror is real

✓ **Correct:** Image in plane mirror is always virtual (cannot be projected)

10.4 Misconception 4

✗ **Wrong:** Minimum mirror length depends on distance from mirror

✓ **Correct:** Minimum mirror length =

$$\frac{1}{2}$$

× height, regardless of distance

11. HKDSE Exam Key Points

11.1 Must Remember

1. **Law of reflection:**

$$i = r$$

2. **Angles measured from normal**, not surface

3. **Plane mirror image:** Virtual, erect, same size, laterally inverted,

$$v = u$$

4. **Regular vs diffuse reflection:** Smooth vs rough surface

5. **Minimum mirror length:**

$$\frac{1}{2}$$

× height

6. **Types of beams:** Parallel, converging, diverging

11.2 Common Question Types

1. **Ray diagrams:** Draw incident and reflected rays

2. **Angle calculations:** Use

$$i = r$$

and geometry

3. **Image location:** Use ray tracing or

$$v = u$$

4. **Mirror length:** Calculate minimum length

5. **Applications:** Identify uses of different mirrors

11.3 Exam Tips

- Always draw normal first
- Measure angles carefully with protractor
- Label all rays: incident, reflected, normal
- State image characteristics clearly
- Show all working in calculations

12. Quick Memory Mnemonics

Law of Reflection:

i equals r, that's the law

Measure from normal, not the floor

Plane Mirror Image:

Virtual, erect, same size too

Laterally inverted, distance equal through

Types of Reflection:

Smooth surface, regular reflection
Rough surface, diffuse direction
Both obey $i = r$ perfection

Minimum Mirror Length:

Half your height, that's the key
To see yourself from head to knee

Study Suggestions:

1. Practice drawing ray diagrams with normal
2. Memorize law of reflection and image characteristics
3. Understand difference between regular and diffuse reflection
4. Practice angle calculations using geometry
5. Learn applications of different types of mirrors